



Department of Mathematics
IIT Hyderabad
MA1110: Calculus I
Fall 2026

Assignment 1

1. Discuss the convergence of the following sequences:

(i) $\left\{ \frac{(-1)^n n}{n^2 + 1} \right\}$ (ii) $\left\{ \frac{3n^2 + \sin n - 4}{2n^2 + 3} \right\}$

(iii) $\left\{ \frac{n+1}{\sqrt{n}} \right\}$ (iv) $\left\{ \left(1 + \frac{1}{n^2}\right)^{n^2} \right\}$

(v) $\left\{ \left(\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2}{n^2}\right) \left(1 + \frac{3}{n^2}\right) \right)^{n^2} \right\}$

2. Determine whether the following sequences are increasing, decreasing, or neither:

(i) $\left\{ \frac{3^n}{3^{n+1}} \right\}$ (ii) $\left\{ \frac{5n-2}{4n+1} \right\}$ (iii) $\left\{ \cos \frac{n\pi}{3} \right\}$

3. Give an example of divergent sequences $\{x_n\}$ and $\{y_n\}$ such that $\{x_n y_n\}$, $\{x_n + y_n\}$ is convergent.

4. If $x_n > 0$ for all $n \in \mathbb{N}$ and $\lim \sqrt[n]{x_n} = l > 0$, show that $\lim \sqrt[n]{(n+1)x_{n+1}} = l$.

5. Examine whether the sequence $\{x_n\}$ where $x_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}$ is convergent or not. If $\{x_n\}$ is convergent, find its limit.

6. Show that the sequence $\{1, \frac{1}{2}, 1, \frac{2}{3}, 1, \frac{3}{4}, \dots\}$ converges to 1.

7. Two sequences $\{x_n\}$ and $\{y_n\}$ are defined by $x_{n+1} = \frac{1}{2}(x_n + y_n)$, $y_{n+1} = \sqrt{(x_n + y_n)}$ for $n \geq 1$ and $x_1 > 0$, $y_1 > 0$. Prove that both sequences converge to a common limit.

8. Give an example of a sequence of rational numbers that converges to an irrational number. Also, give an example of a sequence of irrational numbers that converges to a rational number.

9. If the sequence $\{x_n\}$ is defined as $x_1 = 1$ and $x_{n+1} = \frac{1}{x_n + 2}$ for all $n \in \mathbb{N}$, prove that $\{x_n\}$ is convergent and find its limit.

10. Find the limit of the sequences

(a) $\left\{ \sqrt[n]{\frac{2n!}{n! n!}} \right\}$ (b) $\left\{ \frac{1}{n} \sqrt[n]{(2n+1)(2n+2) \cdots (2n+n)} \right\}$

11. Which of the following statements are correct?

(a) $n \log \left(1 + \frac{1}{n+1} \right) \rightarrow 1$ as $n \rightarrow \infty$ (b) $n \log \left(1 + \frac{1}{n^2} \right) \rightarrow 1$ as $n \rightarrow \infty$

12. Prove that the sequence $\{u_n\}$ defined by the following recurrence relations converges, and find its limit.

(a) If $0 < u_1 < u_2$ and $u_{n+2} = \frac{2u_{n+1} + u_n}{3}$, $n \geq 1$, then show that u_n converges to $\frac{u_1 + 3u_2}{4}$.

(b) If $0 < u_1 < u_2$ and $\frac{2}{u_{n+2}} = \frac{1}{u_{n+1}} + \frac{1}{u_n}$, $n \geq 1$, then show that u_n converges to

$$\frac{3}{\frac{1}{u_1} + \frac{2}{u_2}}.$$

13. If $\sum_{n=1}^{\infty} x_n$ is a convergent series of positive real numbers, prove that

(a) $\sum_{n=1}^{\infty} x_n^2$ is convergent.

(b) $\sum_{n=1}^{\infty} \frac{x_n}{n}$ is convergent.

(c) $\sum_{n=1}^{\infty} x_{2n}$ is convergent.

(d) $\sum_{n=1}^{\infty} \frac{x_n}{1+x_n}$ is convergent.

14. Discuss the convergence or divergence of the following series.

(i) $\sum_{n=1}^{\infty} \frac{n^2 + 1}{(n+3)(n+4)}$

(ii) $\sum_{n=1}^{\infty} \left[\sqrt[3]{n^3 + 1} - n \right]$

(iii) $\sum_{n=1}^{\infty} \frac{\log n}{2n^3 - 1}$

(iv) $\sum_{n=1}^{\infty} \sin^2 \frac{1}{n}$

(v) $\sum_{n=1}^{\infty} n \tan \frac{1}{n^2}$

(vi) $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$

(vii) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} n}{n^2 + 1}$

(viii) $\sum_{n=1}^{\infty} \frac{(-1)^n 3^n}{n!}$

(ix) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} (n!)^2}{(2n)!} 5^n$

(x) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \log(n+1)}{n+1}$

15. Using the fact that $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \log 2$, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1)}$ equals

(A) $1 - 2 \log 2$

(B) $1 + \log 2$

(C) $(\log 2)^2$

(D) $-(\log 2)^2$

16. The series $\sum_{k=0}^{\infty} \frac{k^2 + 3k + 1}{(k+2)!}$ converges to

(A) 1

(B) $\frac{1}{2}$

(C) 2

(D) 3

17. We know that $xe^x = \sum_{n=1}^{\infty} \frac{x^n}{(n-1)!}$. The series $\sum_{n=1}^{\infty} \frac{2^n n^2}{n!}$ converges to

(A) e^2

(B) $2e^2$

(C) $4e^2$

(D) $6e^2$

18. Discuss the convergence of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n\sqrt{n}}$ and $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} n}{n^2 + 1}$.

19. Prove that $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n(\log n)^{1/3}}$ is conditionally convergent.

20. Prove that the series

$$\left(\frac{1}{2}\right)^2 - \left(\frac{1 \cdot 4}{2 \cdot 5}\right)^2 + \left(\frac{1 \cdot 4 \cdot 7}{2 \cdot 5 \cdot 8}\right)^2 - \dots$$

is conditionally convergent.